Eylon Caplan

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Education

Purdue University, West Lafayette, Indiana, USA **(expected)** 2027
Ph.D. in Natural Language Processing, Department of Computer Science **3.8/4.0**

University of Nebraska–Lincoln, Lincoln, Nebraska, USA 2023
B.S. in Computer Science and Mathematics (Minors: Physics, Spanish) **3.99/4.0**

Publications

Splits! A Flexible Dataset for Evaluating a Model's Demographic Social Inference
Eylon Caplan, Tania Chakraborty, and Dan Goldwasser
Under review at COLM 2025

CONCEPTCARVE: Dynamic Realization of Evidence
Eylon Caplan and Dan Goldwasser
ACL 2025 Main Conference

Research Experience

Graduate Researcher, Purdue NLP Lab (Advisor: Dan Goldwasser) 2023–Present

- Conduct research at the intersection of NLP, social reasoning, and retrieval, with a focus on modeling human behavior, beliefs, and values in real-world, unstructured corpora such as Reddit.
- Develop CONCEPTCARVE, a framework that combines LLM reasoning with scalable retrieval to map how abstract social concepts manifest across online communities.
- Create *Splits!*, a 3.6 M–post Reddit dataset with demographic and neutral-topic annotations to benchmark models' ability to generate generalizable theories about group language.
- Develop end-to-end multiprocessing pipelines for large-scale data segmentation and cleaning; design and evaluate retrieval and reranking systems; apply clustering, validation, and LLM-based evaluation at scale.

Industry Experience

Software Engineering Intern, Hudl 2022– 2023
Developed and deployed an OCR model to read in-arena basketball scoreboards from indoor camera feeds and splice digital version onto live video stream.

Teaching and Curriculum Design

Course Developer, Purdue University 2025–Present

- Designed a module and four-part project about the RAG pipeline for the *AI Forge* course. Project included parts teaching model inference, prompting, in-context learning, retrieval, and retrieval augmentation. Also designed an evaluation pipeline of student code on computing cluster.
- Designed assignments and course content for a new course, *Data Structures and Algorithms for AI*. Created four course projects, covering topics like trees, stacks, queues, big data hashing, fuzzy word search, and graphs.

Graduate Teaching Assistant, Purdue University 2024
Led labs and assisted instruction for *Data Structures and Algorithms for Data Science/AI Majors*.

Key Courses

Graduate Level: Advanced Topics in Reasoning with LLMs, NLP, Deep Learning, Reasoning about Programs

Key Course Projects

LLM Feedback for Proofs May 2023–Dec 2023
Tested various methods of injecting feedback from an LLM in order to generate correct symbolic proofs in the Isabelle proof solver for competition math problems. Course project for *Adv. Topics in Reasoning with LLMs*.

Scientific Article RAG for QA Jan 2024–May 2024
Built a RAG framework using ColBERT and a generative graph language model. Compared its ability to use scientific papers to do QA tasks. Course project for *NLP*.

Math Expression Style Transfer Aug 2024–Dec 2024
Developed an LLM BFS algorithm for converting math expressions into various simplified/expanded forms using only examples, with guaranteed equivalence. Course project for *Reasoning about Programs*.

Awards

Corporate Partners Scholarship 2023– 2024
Purdue Science Excellence Scholarship 2023– 2024
Regents Scholarship 2019– 2023
Dean H. and Floreen G. Eastman Scholarship 2021– 2023

Skills

ML/AI, NLP, transformers, RL, multiprocessing, IR, Python, PyTorch, Big Datasets, Benchmarking